Defect-aware graph learning of impurity incorporation energetics and chemical transferability in two-dimensional materials

Yiming Hua[1]

[1] *Wuhan University, Wuhan, China*

(Dated: July 28, 2026)

Machine-learned formation energies can accelerate impurity screening in two-dimensional materials, but random-split accuracy alone does not establish chemical transferability or a reliable deployment domain. We develop a defect-aware graph transformer and evaluate it on a provenance-audited, leakage-controlled subset of the IMP2D database. A paired 2^3 factorial separates the contributions of gated graph readout, defect-environment features, and a pre-normalized, distance-gated local interaction block. The architecture is selected using validation data only and is then tested under random, host-held-out, impurity-held-out, host–impurity-pair-held-out, and predefined compositional-block protocols against periodic SchNet and tuned descriptor baselines. A separate calibration partition supports ensemble uncertainty and split-conformal intervals, while out-of-fold predictions quantify defect-type preference and site-screening regret. This protocol is designed to distinguish interpolation accuracy from chemically meaningful transfer and to state the applicability limits of the learned surrogate without invoking unmatched first-principles calculations as external validation.

I. INTRODUCTION

Point defects and incorporated impurity atoms can alter carrier density, localized electronic states, magnetism, and catalytic activity in two-dimensional (2D) materials. Formation energy is therefore a central quantity in computational defect screening: for a specified host, impurity reservoir, charge state, and structural configuration, it ranks the energetic cost of incorporation. Exhaustive first-principles evaluation remains expensive because the search space grows across host chemistry, impurity identity, incorporation class, and symmetry-distinct site.

The IMP2D database provides a systematic first-principles survey of neutral adsorbate and interstitial impurities in 2D hosts [1]. Its scale makes supervised surrogates possible, and graph neural networks are natural models because they operate directly on periodic atomic structures. ALIGNN demonstrated the value of joint bond and angle representations for crystalline properties [2], while SchNet established continuous-filter message passing from interatomic distances [3]. Defect-specific learning has subsequently developed along several complementary directions. Bulk defect energetics have been treated with defect graph networks, semiconductor graph models, persistent-homology features, and charge-aware predictors [4–7]. For 2D systems, sparse defect representations have been evaluated on the 2DMD data, whereas tree and descriptor models have been applied directly to IMP2D [8–10]. The two IMP2D studies show in particular that data representation and model selection materially affect the reported error. These developments motivate architectures that expose the impurity site while retaining both local coordination and longer-range periodic context.

Benchmark accuracy, however, does not by itself define a useful materials model. A random structure split can place closely related host–impurity chemistries, or even numerically equivalent structures, on both sides of the train–test boundary, a known source of optimistic materials-learning estimates [11]. It also does not answer whether a model transfers to an unseen host or impurity, whether a reported architectural gain survives paired repeats, or whether prediction uncertainty identifies a subset on which screening risk is acceptably low. Structure-based out-of-distribution studies have emphasized that materials models can degrade sharply when test chemistry differs from training chemistry [12, 13]. For defect screening, host and impurity axes must therefore be separated explicitly. Uncertainty studies in atomistic machine learning have likewise shown that ensemble disagreement can provide a useful credibility signal, while calibration and out-of-domain reliability remain distinct requirements [14–16]. A defect-screening surrogate should therefore be judged by its transfer partitions, uncertainty calibration, and decision-level screening consequences rather than by one random-split error.

This work asks three connected questions. First, do defect-centered readout, explicit local-environment features, and a distance-gated local interaction block provide reproducible improvements when tested as a complete paired factorial rather than as sequential additions? Second, how does the selected model compare with periodic SchNet and validation-tuned descriptor baselines when hosts, impurities, host–impurity pairs, or a compositional block are held out? Third, can a separately calibrated ensemble support selective prediction and recover the preferred incorporation class or lowest-energy site with low screening regret?

To answer these questions, we construct a fully auditable protocol around the IMP2D source database. We replay the source filter and formation-energy formula, remove rows whose raw energy components cannot be reconstructed, exclude same-element structures for which the relaxed database cannot assign a permutation-invariant impurity node, merge duplicate structures with both coordinate and invariant-distance fingerprints, and publish immutable split definitions. The resulting 10,224 structure modeling set is evaluated through a paired 2^3 architecture experiment, random and chemistry-grouped cross-validation, and a predefined host–impurity matrix holdout. Architecture and descriptor choices use validation data only. A distinct calibration partition is used for variance scaling and split-conformal intervals, and all materials-facing conclusions are computed from out-of-fold predictions.

The intended contribution is consequently broader than a new point estimate on a familiar random split. It is a test of which defect-aware modules are supported by paired evidence, where chemical transfer fails, and how that failure should constrain use of the surrogate. No unmatched external first-principles calculations are presented as validation; the scope is the accuracy, transferability, calibration, and screening utility that can be established from the audited IMP2D evidence.

II. METHODS

A. Dataset, target provenance, and canonical modeling set

We use neutral impurity structures from the Impurities in Two-Dimensional Materials Database (IMP2D) [1]. The source ASE database contains 17,364 rows. Replaying the data-preparation filter in database order removes 4551 unconverged calculations, 1828 rows with a missing or nonfinite stored formation energy, and 344 rows with $|E_\mathrm{f}| > 20$ eV, leaving 10,641 structures. These structures span 44 hosts, 65 impurity elements, and two incorporation classes (adsorbate and interstitial).

For rows with complete source fields, the target was independently recomputed as

$$E_\mathrm{f} = E_\mathrm{def} - E_\mathrm{host} - \mu_\mathrm{imp}. \tag{1}$$

The recomputed and stored values agree to a maximum absolute numerical difference of 2.3×10^{-13} eV. Four otherwise filtered rows contain a zero sentinel in the stored defective-structure energy and therefore cannot be independently reconstructed from the source components; these rows are excluded from every modeling split.

The released relaxed structures do not retain a persistent identity field for the atom initially inserted by the structure generator. In 349 rows, the impurity element is also a constituent of the host, so multiple identical nuclei match the impurity label. Assigning one of them by array order would break permutation invariance and would make the defect-centered features depend on a nonphysical convention. We therefore require exactly one atom to match the impurity element and exclude all 349 nonunique rows from the formal protocol.

We also prevent symmetry-equivalent structures from crossing split boundaries. Duplicate candidates are identified by the union of two hashes: (i) atomic numbers, cell, and Cartesian coordinates rounded to 10^{-6}, and (ii) a permutation- and translation-invariant spectrum of element-labeled periodic pair distances rounded to 10^{-4} Å. The latter detects numerically equivalent structures even when atom order or coordinate origin differs. Among the remaining eligible structures, connected duplicate groups contain 64 redundant rows; the lowest-index auditable representative is retained. The resulting canonical modeling set contains 10,224 structures. The complete audit, excluded sample indices, database hashes, and immutable split files are distributed with the code.

B. Fixed evaluation partitions

All partitions were generated once from the canonical set and stored as index lists. Five paired random 80/10/10 partitions (assignment seeds 42–46) are used for the 2^3 architecture experiment. A separate fivefold random cross-validation partition provides one out-of-fold prediction for every retained sample. Chemical transfer is evaluated using balanced fivefold group cross-validation in which complete host identities, impurity identities, or host–impurity pairs are held together. In each grouped fold, one group fold is used for testing and the next group fold for validation. We further use a predefined matrix-completion holdout: group-VI dichalcogenide hosts combined with $3d$ impurities form the test block, while group-IV/V hosts combined with $4d$ impurities form the validation block. After the common audit exclusions, this partition contains 9,487 training, 372 validation, and 365 test structures.

Uncertainty quantification uses an independent random partition with 75% for training, 10% for checkpoint selection, 5% for calibration, and 10% for final testing (assignment seed 62). The calibration and test indices are not used to fit network parameters or select checkpoints. All split files cover the full 10,641-row container through mutually disjoint train, validation, test, optional calibration, and exclusion sets; their SHA256 hashes are stored in every run manifest.

Figure 1 summarizes the sequential data audit, canonical chemical coverage, target distribution, and fixed partition geometry.

C. Defect-aware graph transformer

Each periodic supercell is represented by an atomistic graph. We refer to the resulting model as the Defect-Aware Radial Transformer (DART). The initial node representation concatenates nine standardized elemental attributes with a fixed 128-dimensional ct-UAE elemental vector [17]. We reconstruct a 100-element table from the published checkpoint by applying its linear `atom_embed` layer to each one-hot elemental code, $\mathbf{U} = \mathbf{W}^\mathsf{T} + \mathbf{1}\mathbf{b}^\mathsf{T}$. A zero padding row is prepended internally, so atomic numbers 1–100 address rows 0–99 of $\mathbf{U}$ without an index shift. The concatenated representation is projected to 128 hid-

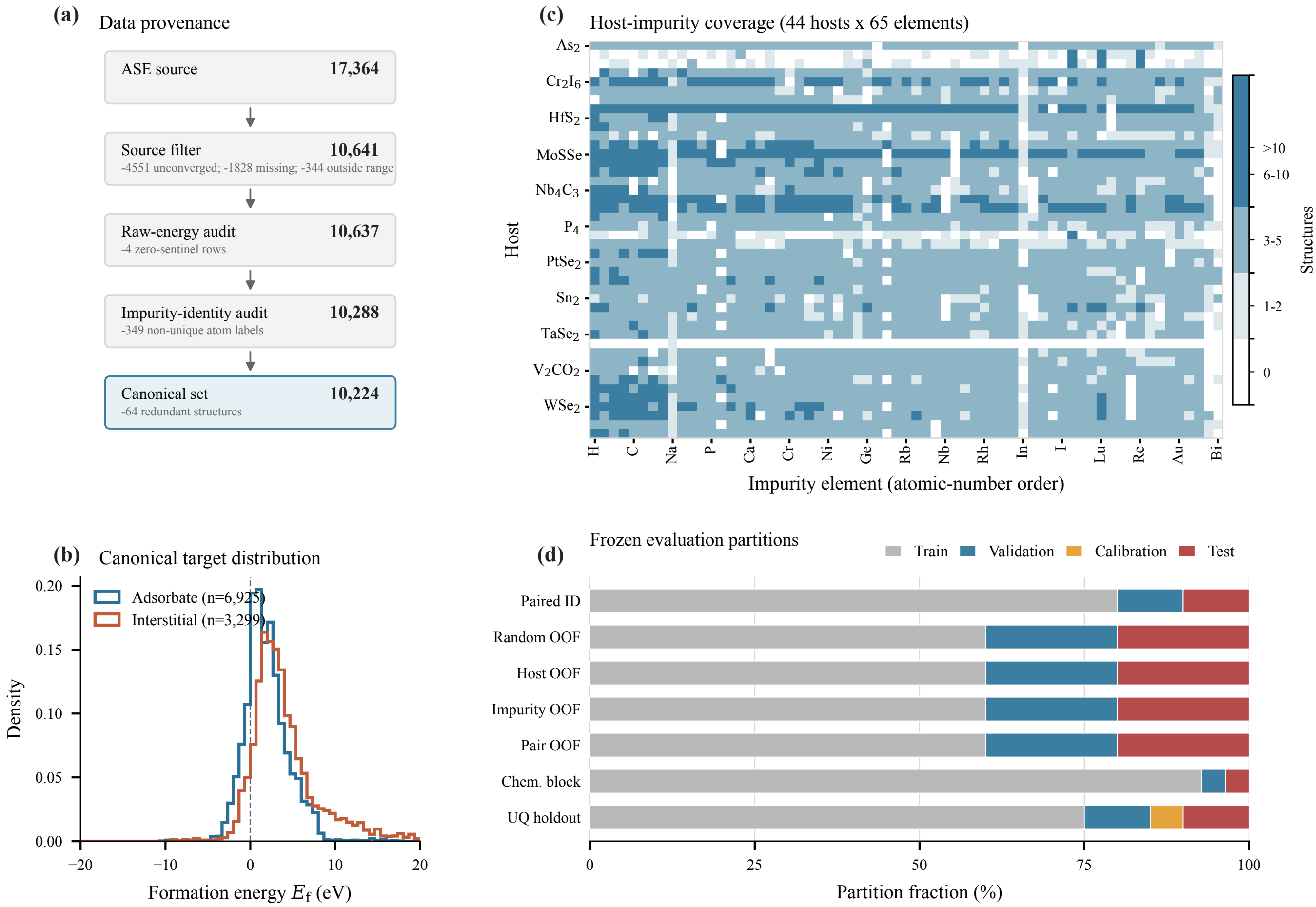

FIG. 1. Data provenance and frozen evaluation protocol. (a) Sequential source filtering, raw-energy reconstruction audit, impurity-identity audit, and structure canonicalization. The final set excludes four rows with zero sentinels in a raw energy component, 349 rows without a unique impurity atom, and 64 redundant structures. (b) Formation-energy distributions for the two incorporation classes in the canonical set. (c) Number of structures for each of 44 hosts and 65 impurity elements; white cells are unobserved combinations. (d) Train, validation, calibration, and test fractions for each evaluation regime. Bars for fivefold protocols show fold averages. All regimes cover the same 10,224 retained structures; the common 417-row exclusion set is omitted from the bars.

den channels, and a learned binary embedding marks the impurity atom. Three local interaction layers aggregate neighbors within 5 Å. Distances are expanded in 32 Gaussian radial basis functions, and three-body messages use a 32-function angular expansion. Two subsequent four-head geometric Transformer blocks apply self-attention with a learned radial bias derived from the periodic pair-distance matrix on a radial grid ending at 12 Å [18]. A two-layer multilayer perceptron maps the pooled graph representation to $\hat{E}_f$. Because every formal row passes the identity audit, the binary impurity embedding always marks exactly one compositionally unique atom; no array-order heuristic is used.

The confirmatory architecture experiment varies three binary modules while holding every other training and model setting fixed:

Gated readout (G).: A learned atom-attention sum and a channelwise maximum are fused by a graph-dependent scalar gate. The off state uses masked mean pooling.

Environment enrichment (E).: Four defect-centered quantities—coordination number, mean and maximum neighbor distance, and the absolute electronegativity contrast to the local neighborhood—are normalized, projected, and added only to the impurity representation. The off state omits this branch.

Pre-norm gated local block (P).: The on state normalizes node states before message construction and applies a learned scalar distance gate to each edge before a residual update. The off state uses the original ungated message and post-residual normalization [19]. We treat this as one composite local-block change and do not attribute its effect to normalization or edge gating separately.

The full 2^3 factorial contains all eight G/E/P combinations on each of the five paired random partitions. Within a partition, all variants share the same initialization seed and training batches. Main effects and interactions are orthogonal factorial contrasts. The architecture promoted to subsequent transfer and uncertainty experiments is the variant with the lowest mean validation MAE; exact ties are resolved by fewer enabled modules and then by binary variant code. Test metrics are not part of the ordering rule.

D. Training protocol

Networks are optimized for 150 epochs with AdamW, a peak learning rate of 5×10^{-4} reached after a ten-epoch linear warmup from 5×10^{-5}, weight decay 10^{-4}, and cosine decay to 10^{-6}. The objective is mean absolute error (MAE), batch size is 64 for DART, and gradient norms are clipped at 5. Targets are standardized using training-partition statistics for optimization and transformed back to eV for all reported errors. The fixed DART regularization recipe is limited to 0.1 dropout and additive Gaussian target noise with standard deviation 0.03 eV; no coordinate or strain augmentation is applied. A host-balanced sampler draws 200 structures per host per epoch; its sequence and the independent target-noise stream are deterministically indexed by model seed and epoch so paired variants see the same batches and noise realizations. Every factorial variant uses the same two fixed input assets. The ct-UAE table described above is a non-trainable buffer. Separately, we pretrained the nine-feature input projection and V1 local interaction stack for 30 epochs. We first drew 20,000 records uniformly without replacement from JARVIS-DFT `dft_3d` with seed 42; conversion and graph-validity filters retained 19,902 pristine three-dimensional crystals [20]. Pretraining used formation energy per atom as the source target, an 80/10/10 random partition, and seed 42. No IMP2D target is used in this stage. An explicit chemistry-overlap audit found that 26 of the 44 IMP2D host labels (25 of 42 distinct reduced host formulas) occur among 69 JARVIS source records, and that all 65 IMP2D impurity elements occur somewhere in the source-task structures. Formula overlap does not establish structural identity, but it means that the grouped partitions withhold IMP2D formation-energy supervision rather than all prior source-task exposure to the same reduced host formula or impurity element.

The resulting checkpoint contributes its nine-feature input-projection block and bias together with all 51 shape-matched V1 local-message tensors. The 128 input-projection columns associated with the fixed ct-UAE vectors retain their seeded initialization; for P-enabled variants, the 12 new edge-gate tensors do likewise. All trainable model parameters are then optimized on the assigned IMP2D training partition. Both input assets are verified against SHA-256 digests, and each run manifest records the exact copied and seeded parameter sets.

The checkpoint with minimum validation MAE is evaluated once on its test partition. Each run records the code commit and dirty state, resolved configuration, command, Python/PyTorch/CUDA environment, data and split hashes, seed, partition indices, epoch history, checkpoint, validation and test predictions, and calibration predictions when the split defines them. SHA-256 digests bind the metrics, checkpoint, split-index, and prediction files to that manifest. Before aggregation, collectors verify partition membership and independently recompute all reported regression metrics from the stored targets and predictions. Interrupted jobs resume from complete optimizer, scheduler, model, and random-number-generator state.

E. Baselines

Two baseline classes use exactly the same canonical partitions. First, an 80-dimensional deterministic descriptor combines elemental-property statistics, impurity–host contrasts, cell geometry, nearest-neighbor distances, coordination counts, and one-hot site and defect labels. We fit a training-target mean predictor, ridge regression, random forests, histogram gradient boosting, and LightGBM [21]. These model families also permit direct comparison with earlier IMP2D descriptor studies [9, 10]. Hyperparameters within each family are selected by validation MAE only; the best descriptor family used in each reported regime is likewise chosen by mean validation MAE across folds.

Second, a periodic SchNet comparator [3] uses 128 hidden channels, six interaction blocks, 50 Gaussian functions, and a 5 Åcutoff. It uses batch size 32 but otherwise follows the same target normalization, 150-epoch AdamW schedule, host balancing, gradient clipping, and target-noise protocol as DART; it uses a conventional learned atomic-number embedding and no DART pretraining. Random and host–impurity-pair fivefold evaluation uses one prespecified model seed per fold (242 for DART and 342 for SchNet). Host-held-out, impurity-held-out, and chemistry-block evaluation uses three independently prespecified seeds (242–244 for DART and 342–344 for SchNet). Predictions are averaged within each split before pooled out-of-fold metrics are calculated. Consequently, DART–SchNet contrasts compare the complete prespecified learning recipes, including their different input representations and initialization; they do not isolate a pure backbone-architecture effect. For the same reason, "host held out" and "impurity held out" refer specifically to IMP2D target supervision.

F. Held-out uncertainty calibration and screening analysis

Five independently initialized members of the validation-selected DART architecture are trained on the dedicated uncertainty split, following the deep-ensemble construction and its atomistic use as a model-credibility signal [14, 22]. Ensemble mean and sample standard deviation provide the point prediction and a raw model-disagreement proxy. A fixed-seed random permutation (seed 6201) assigns the dedicated calibration rows to two subsets without using target labels. One subset fits a nonnegative scale and floor for

$$\sigma_i = \sqrt{(as_i)^2 + b^2} \qquad (2)$$

by Gaussian negative log likelihood; the other subset estimates finite-sample split-conformal quantiles of $|E_{\mathrm{f},i} - \hat{E}_{\mathrm{f},i}|/\sigma_i$ [15, 23]. Under exchangeability, the resulting intervals have finite-sample marginal coverage at each prespecified level; this guarantee does not imply conditional coverage for every host or impurity group. Test labels do not enter either calibration step, and test metrics are computed only after the scale, floor, and interval quantiles are fixed.

For materials-facing analysis, the five host–impurity-pair folds yield one out-of-fold prediction per retained structure. Within each host–impurity pair we compare the minimum predicted and reference energies of adsorbate and interstitial configurations, identify the predicted lowest-energy site, and report preference accuracy and screening regret. Exact-site metrics require at least two candidate sites, whereas top-two accuracy requires at least three; reference minima within 10^{-8} eV are treated as tied, and tied adsorbate–interstitial pairs are excluded from binary preference accuracy. These are predictive associations within IMP2D, not causal mechanism assignments or external DFT validation.

III. EVALUATION PROTOCOL

A. Metrics and statistical analysis

We report MAE, root-mean-square error (RMSE), signed bias, Spearman rank correlation, and R^2. Host- and impurity-macro MAE give each chemical group equal weight, independent of its number of structures. Cross-validation predictions are concatenated only when test folds are disjoint; model seeds are averaged within a fold before concatenation. Low-energy behavior is assessed by MAE for $E_{\mathrm{f}} \leq 0$, MAE within the true lowest-energy decile of each pooled test regime, and recall of that decile by the predicted lowest-energy decile. These strata are evaluation summaries and are not used for architecture or hyperparameter selection.

For the architecture factorial, each effect is evaluated as the difference between the average metric at positive and negative levels of the associated orthogonal contrast within each paired repeat. Ninety-five-percent intervals are obtained by nonparametric bootstrap over the five repeat-level contrasts. Because only five paired repeats were prespecified, the intervals are accompanied by repeat-level sign consistency and interpreted as descriptive uncertainty summaries rather than large-sample hypothesis tests. For model comparisons on aligned out-of-fold predictions, we use 10,000 paired bootstrap resamples. The resampling unit matches the held-out unit: individual structures for random cross-validation, complete hosts for host-held-out evaluation, complete impurities for impurity-held-out evaluation, and host–impurity pairs for pair-held-out and compositional-block evaluation. Within each resampled chemical unit all associated structures remain together. Screening intervals use 20,000 host–impurity-pair cluster-bootstrap resamples. Adsorbate and interstitial site decisions from the same pair remain together, while point estimates average all eligible decision sets. An interval excluding zero is treated as evidence for a directional difference; otherwise the result is reported as inconclusive rather than equivalent.

Uncertainty evaluation includes empirical interval coverage and mean width at 50%, 80%, 90%, and 95% nominal coverage, Gaussian negative log likelihood, the continuous ranked probability score, Spearman correlation between uncertainty and absolute error, and risk–coverage curves. The area under the risk–coverage curve (AURC) is accompanied by its excess over oracle ordering [24]. All calibration choices use only the dedicated calibration partition.

No new first-principles calculations are used as external validation in this study. Previously generated quantum-chemistry files in the project archive are not matched to the canonical IMP2D targets and therefore are excluded from the evidence supporting predictive accuracy or materials discovery claims.

IV. RESULTS

Canonical result assets are pending completion of the predefined GPU experiment chain. Historical exploratory scores and unmatched first-principles calculations are intentionally excluded from this draft.

V. DISCUSSION

Discussion will be populated only from the canonical result assets after all prespecified non-DFT experiments finish.

VI. CONCLUSION

The numerical conclusion is withheld until canonical result assets pass the complete provenance and coverage contract.

DATA AND CODE AVAILABILITY

The source IMP2D data are available through the repository associated with Ref. [1]. The exact sample audit, immutable split definitions, resolved experiment configurations, analysis scripts, and machine-readable result bundles for this study are available at `https://github.com/chimeraHHH/2d-defect-dast`. A release-specific commit and archival identifier will be added before publication.

ACKNOWLEDGMENTS

This work was supported by the National College Students Innovation and Entrepreneurship Training Program (Key Support Area Project). Computations were performed on an NVIDIA L40S cluster at Wuhan University.

[1] J. Davidsson, F. Bertoldo, K. S. Thygesen, and R. Armiento, Absorption versus adsorption: high-throughput computation of impurities in 2D materials, npj 2D Materials and Applications **7**, 26 (2023).

[2] K. Choudhary and B. DeCost, Atomistic line graph neural network for improved materials property predictions, npj Computational Materials **7**, 185 (2021).

[3] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, SchNet – a deep learning architecture for molecules and materials, The Journal of Chemical Physics **148**, 241722 (2018).

[4] M. D. Witman, A. Goyal, T. Ogitsu, A. H. McDaniel, and S. Lany, Defect graph neural networks for materials discovery in high-temperature clean-energy applications, Nature Computational Science **3**, 675 (2023).

[5] M. H. Rahman, P. Gollapalli, P. Manganaris, S. K. Yadav, G. Pilania, B. DeCost, K. Choudhary, and A. Mannodi-Kanakkithodi, Accelerating defect predictions in semiconductors using graph neural networks, APL Machine Learning **2**, 016122 (2024).

[6] Z. Fang and Q. Yan, Leveraging persistent homology features for accurate defect formation energy predictions via graph neural networks, Chemistry of Materials **37**, 1531 (2025).

[7] S. Kiyohara, C. Shibui, S. Bae, and Y. Kumagai, Machine-learning prediction of charged-defect formation energies from crystal structures, Physical Review Letters **135**, 246101 (2025).

[8] N. Kazeev, A. R. Al-Maeeni, I. Romanov, M. Faleev, R. Lukin, A. Tormasov, A. H. Castro Neto, K. S. Novoselov, P. Huang, and A. Ustyuzhanin, Sparse representation for machine learning the properties of defects in 2D materials, npj Computational Materials **9**, 113 (2023).

[9] A. Kesorn, R. Hunkao, C. Na Talang, C. Cholsuk, A. Sinsarp, T. Vogl, S. Suwanna, and S. Yuma, Formation energy prediction of neutral single-atom impurities in 2D materials using tree-based machine learning, Machine Learning: Science and Technology **5**, 035039 (2024).

[10] A. El Alouani, M. Al Khalfioui, A. Michon, S. Vézian, P. Boucaud, and M. T. Dau, Defect formation energy of impurities in 2D materials: How does data engineering shape machine learning model selection?, Materials Today Physics **61**, 102006 (2026).

[11] Q. Li, N. Fu, S. S. Omee, and J. Hu, MD-HIT: Machine learning for material property prediction with dataset redundancy control, npj Computational Materials **10**, 245 (2024).

[12] S. S. Omee, N. Fu, R. Dong, M. Hu, and J. Hu, Structure-based out-of-distribution (OOD) materials property prediction: a benchmark study, npj Computational Materials **10**, 144 (2024).

[13] K. Li, A. N. Rubungo, X. Lei, D. Persaud, K. Choudhary, B. DeCost, A. B. Dieng, and J. Hattrick-Simpers, Probing out-of-distribution generalization in machine learning for materials, Communications Materials **6**, 9 (2025).

[14] A. A. Peterson, R. Christensen, and A. Khorshidi, Addressing uncertainty in atomistic machine learning, Physical Chemistry Chemical Physics **19**, 10978 (2017).

[15] Y. Hu, J. Musielewicz, Z. W. Ulissi, and A. J. Medford, Robust and scalable uncertainty estimation with conformal prediction for machine-learned interatomic potentials, Machine Learning: Science and Technology **3**, 045028 (2022).

[16] F. Grasselli, S. Chong, V. Kapil, S. Bonfanti, and K. Rossi, Uncertainty in the era of machine learning for atomistic modeling, Digital Discovery **4**, 2654 (2025).

[17] L. Jin, Z. Du, L. Shu, Y. Cen, Y. Xu, Y. Mei, and H. Zhang, Transformer-generated atomic embeddings to enhance prediction accuracy of crystal properties with machine learning, Nature Communications **16**, 1210 (2025).

[18] T. Taniai, R. Igarashi, Y. Suzuki, N. Chiba, K. Saito, Y. Ushiku, and K. Ono, Crystalformer: Infinitely connected attention for periodic structure encoding, in *International Conference on Learning Representations (ICLR)* (2024).

[19] R. Xiong, Y. Yang, D. He, K. Zheng, S. Zheng, C. Xing, H. Zhang, Y. Lan, L. Wang, and T.-Y. Liu, On layer normalization in the transformer architecture, in *International Conference on Machine Learning (ICML)* (2020).

[20] K. Choudhary, K. F. Garrity, A. C. E. Reid, B. DeCost, A. J. Biacchi, A. R. Hight Walker, Z. Trautt, J. Hattrick-Simpers, A. G. Kusne, A. Centrone, A. Davydov, J. Jiang, R. Pachter, G. Cheon, E. Reed, A. Agrawal, X. Qian, V. Sharma, H. Zhuang, S. V. Kalinin, B. G. Sumpter, G. Pilania, P. Acar, S. Mandal, K. Haule, D. Vanderbilt, K. Rabe, and F. Tavazza, The joint au-

tomated repository for various integrated simulations (JARVIS) for data-driven materials design, npj Computational Materials **6**, 173 (2020).

[21] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).

[22] B. Lakshminarayanan, A. Pritzel, and C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).

[23] A. N. Angelopoulos and S. Bates, Conformal prediction: A gentle introduction, Foundations and Trends in Machine Learning **16**, 494 (2023).

[24] R. El-Yaniv and Y. Wiener, On the foundations of noise-free selective classification, Journal of Machine Learning Research **11**, 1605 (2010).